Theoretical Research Report: EABC-Proof-Theory (Version 19)

Research Collaboration

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Abstract

This document establishes formal-logical and algebraic components of the Euler-Adic-Borel-Cantor (EABC) research program. We formulate the orthogonal Pythagorean ANOVA decomposition of the Hurwitz–Hoffbauer defect, derive its exact representation as an unnormalized window-integrated algebraic two-point functional, prove that the uniform asymptotic frequency set over the four-symbol EABC alphabet is $\mathbf{\Pi}_3^0$-complete, and analyze a specified eight-state isotropic reward-controlled Markov kernel. For this kernel the limiting entropy is positive, so deterministic zero-entropy crystallization is excluded.

1 Pythagorean Orthogonal Decomposition and Variance Splitting

Let $\eta_{\min} := (1,1,1,1)$ denote the minimal EABC reference pattern, and let $P = (h_E, h_A, h_B, h_C)$ be the observed Hoffbauer bracket. Define

$$\delta := P - \eta_{\min} = (\delta_E, \delta_A, \delta_B, \delta_C), \qquad \delta_X := h_X - 1,$$

and

$$K := \langle \delta, \mathbf{1}_4 \rangle = \sum_{X \in \{E,A,B,C\}} \delta_X, \qquad \mathbf{1}_4 := (1,1,1,1).$$

Since $\|\mathbf{1}_4\|_2^2 = 4$, the space $\mathbb{R}^4$ decomposes as

$$\mathbb{R}^4 = \operatorname{span}\{\mathbf{1}_4\} \overset{\perp}{\oplus} \mathbf{1}_4^{\perp}, \qquad \delta = \underbrace{\frac{K}{4}\mathbf{1}_4}_{\delta_\parallel} + \underbrace{\left(\delta - \frac{K}{4}\mathbf{1}_4\right)}_{\delta_\perp}.$$

By construction, $\langle \delta_\perp, \mathbf{1}_4 \rangle = 0$.

Define the unminimized direct defect by

$$D^0_{\mathrm{HH}}(P) := \|\delta\|_2^2 = \sum_{X \in \{E,A,B,C\}} \delta_X^2.$$

The Pythagorean theorem gives

$$\boxed{D^0_{\mathrm{HH}}(P) = \underbrace{\frac{K^2}{4}}_{\text{radial quadratic contribution}} + \underbrace{D^{\perp}_{\mathrm{HH}}(P)}_{\text{internal EABC asymmetry}}}$$

with

$$D^{\perp}_{\mathrm{HH}}(P) := \|\delta_\perp\|_2^2 = \sum_X \left(\delta_X - \frac{K}{4}\right)^2$$
$$= \sum_X \delta_X^2 - \frac{1}{4}\left(\sum_X \delta_X\right)^2 = 4\ \mathrm{Var}_{\mathrm{pop}}(\delta_E, \delta_A, \delta_B, \delta_C).$$

1.1 Representation as an Integrated Two-Point Functional

Let I be a finite window of EABC prime classes. Put

$$N := |I|, \qquad h_X := \#\{t \in I : c_t = X\}.$$

Then $\sum_X h_X = N$ and $K = \sum_X (h_X - 1) = N - 4$. For the centered simplex embedding

$$v(c_t) := e_{c_t} - \frac{1}{4}\mathbf{1}_4 \in \mathbf{1}_4^{\perp}$$

one has

$$\sum_{t\in I} v(c_t) = \left(h_E - \frac{N}{4}, h_A - \frac{N}{4}, h_B - \frac{N}{4}, h_C - \frac{N}{4}\right) = \delta_{\perp}.$$

Consequently,

$$\boxed{\begin{aligned} D^{\perp}_{\mathrm{HH}}(I) &= \left\| \sum_{t\in I} v(c_t) \right\|_2^2 \\ &= \sum_{r,s\in I} \langle v(c_r), v(c_s) \rangle \\ &= \sum_{X\in\{E,A,B,C\}} h_X^2 - \frac{N^2}{4}. \end{aligned}}$$

Thus $D^{\perp}_{\mathrm{HH}}$ is an unnormalized, window-integrated algebraic two-point functional. No physical or thermodynamic assumptions are required for this identity.

1.2 Range of the Relative Anisotropy Index

For $D^0_{\mathrm{HH}}(P) > 0$, define

$$R_{\perp}(P) := \frac{D^{\perp}_{\mathrm{HH}}(P)}{D^0_{\mathrm{HH}}(P)} = 1 - \frac{K^2}{4D^0_{\mathrm{HH}}(P)}.$$

It is invariant under nonzero rescaling $\delta \mapsto \lambda\delta$.

1. In the ambient defect space $\mathbb{R}^4$, Cauchy–Schwarz gives

   $$0 \le R_{\perp} \le 1,$$

   with

   $$R_{\perp} = 0 \iff \delta \in \mathrm{span}\{\mathbf{1}_4\}, \qquad R_{\perp} = 1 \iff K = 0 \text{ and } \delta \neq 0.$$

2. In the positive excess cone $\delta_X \ge 0$, nonnegativity of the cross terms and Cauchy–Schwarz give

   $$D^0_{\mathrm{HH}}(P) \le K^2 \le 4D^0_{\mathrm{HH}}(P), \qquad \boxed{0 \le R_{\perp} \le \frac{3}{4}}.$$

   Here $R_{\perp} = 0$ if and only if all four components of δ are equal. For a nonzero integer-valued radial defect this implies $4 \mid K$. The equality $R_{\perp} = 3/4$ holds if and only if exactly one component is nonzero; up to permutation, $\delta = (K, 0, 0, 0)$ with $K > 0$.

2 Descriptive Complexity and Topological Classification

Let $\mathcal{C} := \{E, A, B, C\}$ and let $Y := \mathcal{C}^{\mathbb{N}}$ carry the product topology. Define

$$A_{\text{freq}} := \left\{ \omega \in Y : \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{X\}}(\omega_i) = \frac{1}{4} \text{ for every } X \in \mathcal{C} \right\}.$$

We use the standard prototype

$$\mathcal{Z} := \left\{ \alpha \in 2^{\mathbb{N}} : \lim_{n\to\infty} \frac{1}{n} \sum_{i=1}^{n} \alpha_i = 0 \right\},$$

which is $\mathbf{\Pi}^0_3$-complete; see [1].

Theorem 1. *The uniform asymptotic frequency set A_{freq} is a $\mathbf{\Pi}^0_3$-complete (equivalently, $F_{\sigma\delta}$) subset of $\mathcal{C}^{\mathbb{N}}$.*

Proof. For the upper bound, for $X \in \mathcal{C}$, $m \geq 1$, and $N \geq 1$, let

$$F_{X,m,N} := \bigcap_{n\geq N} \left\{ \omega : \left| \frac{1}{n} \sum_{i=1}^{n} \mathbf{1}_{\{X\}}(\omega_i) - \frac{1}{4} \right| \leq \frac{1}{m} \right\}.$$

Each set inside the intersection depends on a finite prefix and is clopen; hence $F_{X,m,N}$ is closed. Moreover,

$$A_{\text{freq}} = \bigcap_{X\in\mathcal{C}} \bigcap_{m=1}^{\infty} \bigcup_{N=1}^{\infty} F_{X,m,N},$$

so $A_{\text{freq}} \in \mathbf{\Pi}^0_3$.

For hardness, define a continuous map $f : 2^{\mathbb{N}} \to \mathcal{C}^{\mathbb{N}}$ using blocks of constant length four. Let

$$H := (E, A, B, C), \qquad S := (E, E, E, E),$$

and set

$$f(\alpha) := B_1 B_2 B_3 \cdots, \qquad B_k := \begin{cases} H, & \alpha_k = 0, \\ S, & \alpha_k = 1. \end{cases}$$

Every output prefix depends on only finitely many input bits, so f is continuous. Write $d_n := n^{-1} \sum_{k=1}^{n} \alpha_k$. At the end of the nth block,

$$\text{freq}_{4n}(X; f(\alpha)) = \frac{1 - d_n}{4} \quad (X \in \{A, B, C\}),$$

whereas

$$\text{freq}_{4n}(E; f(\alpha)) = \frac{1}{4} + \frac{3}{4} d_n.$$

The discrepancy between an arbitrary prefix and the preceding block endpoint is at most $4/N$, which tends to zero. Hence

$$\alpha \in \mathcal{Z} \iff d_n \to 0 \iff f(\alpha) \in A_{\text{freq}}.$$

Thus $\mathcal{Z} \leq_{\text{W}} A_{\text{freq}}$, proving $\mathbf{\Pi}^0_3$-hardness and therefore completeness. □

3 Markov-Kernel Analysis and Entropy Limits

Let $S_8 := \mathcal{C} \times \{+, -\}$, so $|S_8| = 8$. This is a specified model state space; no arithmetic identification beyond this product definition is asserted. For fixed $T_{\text{eff}} > 0$ and scale parameter $\rho > 0$, put

$$\beta_\rho := \frac{\rho^2}{T_{\text{eff}}}, \qquad E_{ij}(\rho) := -\rho^2 \kappa_{ij}, \qquad \kappa_{ii} = 0, \qquad \kappa_{ij} = 1 \quad (i \neq j).$$

The associated Gibbs kernel is

$$T_{ij}(\rho) = \frac{\exp(\beta_\rho \kappa_{ij})}{\sum_{k=1}^{8} \exp(\beta_\rho \kappa_{ik})}.$$

As $\rho \to \infty$,

$$T(\rho) \longrightarrow T_\infty := \frac{1}{7}(J_8 - I_8).$$

3.1 Entropy and Information Limits

The formal high-temperature limit $\beta_\rho \to 0$ has row entropy

$$H_{\text{high}} = \log_2 8 = 3 \text{ bits}.$$

In the low-effective-temperature limit $\beta_\rho \to \infty$,

$$H_\infty = \log_2 7 \approx 2.80735 \text{ bits}, \qquad \Delta H = \log_2 \frac{8}{7} \approx 0.192645 \text{ bits}.$$

Under the uniform stationary distribution of the limiting chain,

$$I(X_n; X_{n+1}) = \log_2 \frac{8}{7}.$$

This mutual information records exactly the exclusion of the current state from the seven admissible successors.

3.2 Spectral Characterization

The spectrum is

$$\operatorname{spec}(T_\infty) = \left\{ 1, \left(-\frac{1}{7}\right)^{[7]} \right\}.$$

With $\Pi := J_8/8$,

$$T_\infty^n = \Pi + \left(-\frac{1}{7}\right)^n (I_8 - \Pi).$$

Thus every initial distribution converges geometrically to the uniform stationary distribution. The limiting kernel describes isotropic self-avoidance, not deterministic crystallization.

4 Epistemic Status Summary

Component	Status
Pythagorean defect decomposition	[T] Proved
Integrated two-point functional identity	[T] Proved
Anisotropy bounds in the stated domains	[T] Proved
Uniform frequency set is $\mathbf{\Pi}^0_3$-complete	[T] Proved, using the cited prototype theorem
Specified Markov limit $T_\infty = (J_8 - I_8)/7$	[T] Proved
Deterministic zero-entropy limit for this kernel	[F] Refuted
Application of the eight-state kernel to prime dynamics	[H] Open

References

[1] H. Ki and T. Linton, *Normal numbers and subsets of* $\mathbb{N}$ *with given densities*, Fundamenta Mathematicae **144** (1994), 163–179.